Problem Statement: Collaborative Digital Art Layer Manager

Design and implement a Collaborative Digital Art Layer Manager using a doubly linked list to handle overlapping design layers in an image editor. Each layer is a node holding its name, pixel data reference, opacity, visibility, and active status. The system must support inserting a new layer at the top or bottom of the stack, deleting inactive layers while correctly relinking their neighbours, and reordering layers through pointer manipulation without copying layer data. Traversal from bottom to top must composite the visible layers in order so that upper layers overlay lower ones, producing the final canvas. Boundary cases such as empty lists, single-layer lists, and head or tail deletion must be handled correctly. The structure must preserve its invariants, with correct prev and next links at every node and null ends at head and tail. Structural operations should run in constant time, while deletion-by-scan and rendering run in linear time in the number of layers. Deliverables include the source code, a driver program demonstrating each operation, test cases for edge conditions, and a short report explaining the design and complexity. Evaluation will focus on pointer correctness, code clarity, complexity analysis, test coverage, and the accuracy of the final rendered layer order.